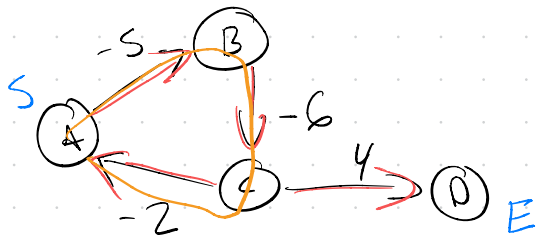
Provide an example where Dijkstra's algorithm fails to find the shortest path in a graph even when the graph contains no negative weight cycles. Include the following:

1) The graph
2) The start & end vertices for the counter examples
3) The distance computed by Dijkstra's algorithm vs. the actual shortest distance.

Dijkstra's algo. pseudocode:

1) Initialize
   - Set start node dist. to 0
   - Set all others to inf.
   - Add start to the min pq.
2) Loop until our pq.
3) Relax neighbor distances
4) Add them to the queue

Negative Weight Cycle:



$C = -\infty + 4 = -\infty$

Min PQ:

~~A~~ = 0

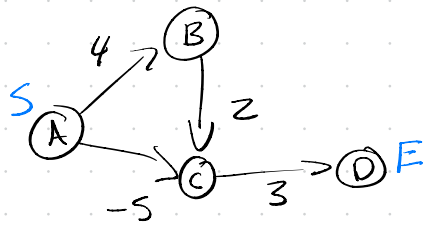
~~B~~ = ~~∞~~ -> -5

~~C~~ = ∞ -> -11

~~D~~ = ∞ -> -7

A = -13

Counter Example:



Min PQ:

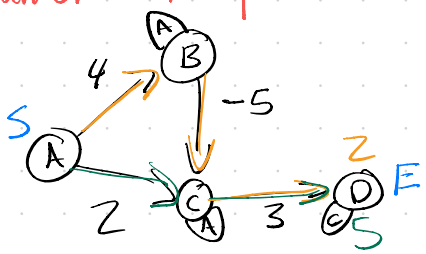
~~A~~ = 0

~~B~~ = ~~-∞~~ -> 4

~~C~~ = ~~-∞~~ -> -5

~~D~~ = -∞ -> -2

Counter Example:



Min PQ:

~~A~~ = 0

~~B~~ = ~~-∞~~ -> 4

~~C~~ = ~~-∞~~ -> ~~2~~ -> -1

~~D~~ = ~~-∞~~ -> 5

Dijkstra's: 5 as D's shortest path cost

Actual: 2 is the actual shortest path